# CSE 270 Sports Analytics

## Predicting with models

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# Poisson regression

## Predicting game outcome with Poisson regression

# Poisson regression

Libraries used

```
library(dplyr)
library(VGAM)
library(ggplot2)
library(SportsAnalytics270)
library(BradleyTerry2)
```

# Poisson regression

First we are going to test the hypothesis that Home goals and Away goals are independent.

We will use a $\chi^2$ test of independence of two categorical variables

$$H_0 : \text{Two variables are independent}$$

$$H_1 : \text{Two variables are NOT independent}$$

Calculate the p-value and compare with significance level of $\alpha$, if p-value $< \alpha$ reject the NULL hypothesis

# Poisson regression

First we are going to create categoric variables for Home and Away team goals in a way that more than 3 goals will be combined together

```
data("f_data_sm")
df = f_data_sm %>% mutate(FTAG = ifelse(FTAG>=4, "4+", FTAG),
                          FTHG = ifelse(FTHG >=4, "4+", FTHG))
```

# Poisson regression

Recall that $\chi^2$ test is using Expected and Observed frequencies as an input. Thus we will need to provide a table with joint frequencies of Home and away team goals

```
table(Away=df$FTAG, Home = df$FTHG)
```

```
##      Home
## Away     0     1     2     3    4+
##    0  8654 11525  8965  5002  3542
##    1  7848 13120  9688  5042  3263
##    2  5049  6974  5598  2691  1622
##    3  2315  3016  1982  1081   581
##    4+ 1368  1490   947   399   190
```

# Poisson regression

We do the test for the last season

- We use function chisq.test
- We specify the table with Observed Frequencies
- And specify simulate.p.values = TRUE: This will add robustness to our test
- We do the test for each country and extract p-values of the test

# Poisson regression

```
df %>% filter(SEASON==2026) %>% group_by(COUNTRY) %>%
  summarise(pvalue=chisq.test(table(FTAG, FTHG),
                              simulate.p.value=T)$p.value)
```

```
## # A tibble: 11 x 2
##    COUNTRY       pvalue
##    <chr>          <dbl>
##  1 Belgium      0.939
##  2 England      0.233
##  3 France       0.844
##  4 Germany      0.00300
##  5 Greece       0.586
##  6 Italy        0.285
##  7 Netherlands  0.602
##  8 Portugal     0.000500
##  9 Scotland     0.796
## 10 Spain        0.336
## 11 Turkey       0.410
```

In general, the hypothesis of independence is not rejected, at $\alpha = 0.05$ , Thus they are independent

# Poisson regression

- Poisson regression is used when the dependent variable is a count data, i.e. y=0,1,2,3. . . , thus we assume that the dependent variable has Poisson distribution
- Independent variables can be both numeric and categorical
- Poisson regression is a part of Generalized Linear Regression (GLM) models family

# Poisson regression: the general definition

| Common Name | General Model Form | Link Function | Distribution | Notes |
|---|---|---|---|---|
| Linear Regression | $E(Y\|X) = \beta_0 + \beta_1 X_1 + \beta_p X_p$ | Identity | Normal (Gaussian) | Continuous regressand |
| Logistic Regression | $ln\left[\frac{E(Y\|X)}{1 - E(Y\|X)}\right] = \beta_0 + \beta_1 X_1 + \beta_p X_p$ | Logit | Bernoulli (Binomial) | Dichotomous regressand |
| Poisson Regression | $ln[E(Y\|X)] = \ln(t) + \beta_0 + \beta_1 X_1 + \beta_p X_p$ | Log | Poisson | Regressand is count over period of time |
| Cox Proportional Hazards Regression | $ln[h(t\|X)] = ln[h_0(t)] + \beta_1 X_1 + \beta_p X_p$ | Identity | No Distribution Assumed | Censored data |

# Poisson regression: the general definition

In this chapter we are fitting the model of the following form

$$log(\lambda) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ...$$

Where $\lambda$ is the mean of Poisson distribution and $X_1, ..X_n$ are the independent variables

With the estimate of the $\lambda$ we can generate Poisson distribution and calculate the probabilities for the number of the goals

# Preparing data for the Poisson regression model

The initial data for England, season 2025-2026 Making Predictions for English Premier League

```
data(f_data_sm)
England = f_data_sm %>%
            filter(SEASON==2026, COUNTRY=="England") %>%
            select(HOMETEAM, AWAYTEAM, FTHG, FTAG)

head(England, n=5)
```

```
##      HOMETEAM    AWAYTEAM FTHG FTAG
## 1   Liverpool Bournemouth    4    2
## 2 Aston Villa   Newcastle    0    0
## 3    Brighton      Fulham    1    1
## 4  Sunderland    West Ham    3    0
## 5   Tottenham     Burnley    3    0
```

```
dim(England)
```

```
## [1] 380   4
```

# Preparing data for the Poisson regression model

The data need to be in the following format: 2 rows for each game and the indicator showing if the team is a home team

```
eng1=data.frame(England[,c("HOMETEAM", "AWAYTEAM", "FTHG")], Home=1)
eng2=data.frame(England[,c("AWAYTEAM", "HOMETEAM", "FTAG")], Home=0)
colnames(eng1)=c("Team", "Opponent", "Goal", "Home")
colnames(eng2)=c("Team", "Opponent","Goal", "Home")
England2=rbind(eng1, eng2)
head(England2)
```

```
##          Team    Opponent Goal Home
## 1   Liverpool Bournemouth    4    1
## 2 Aston Villa   Newcastle    0    1
## 3    Brighton      Fulham    1    1
## 4  Sunderland    West Ham    3    1
## 5   Tottenham     Burnley    3    1
## 6      Wolves    Man City    0    1
```

# Preparing data for the Poisson regression model

Now let's have a look the game: Tottenham vs Man United

- Tottenham was the home team, and the game ended with 2-0

```
(tottenham_vs_mu <- England %>%
  filter(HOMETEAM == "Tottenham", AWAYTEAM == "Man United"))
```

```
##     HOMETEAM    AWAYTEAM FTHG FTAG
## 1 Tottenham Man United    2    2
```

```
home_team_index <- which(England$HOMETEAM == "Tottenham" &
                           England$AWAYTEAM == "Man United")
```

# Preparing data for the Poisson regression model

- Let's look at our new dataset

```
England2[home_team_index,]
```

```
##          Team   Opponent Goal Home
## 101 Tottenham Man United    2    1
```

```
England2[home_team_index + nrow(England),]
```

```
##           Team  Opponent Goal Home
## 481 Man United Tottenham    2    0
```

The number of the rows in the England2 dataset is twice of the original one

```
dim(England)
```

```
## [1] 380   4
```

```
dim(England2)
```

```
## [1] 760   4
```

# The model

```
model=glm(Goal~Team+Opponent+Home, data=England2, family=poisson(link=log))
coefficients(model)
```

```
##           (Intercept)        TeamAston Villa        TeamBournemouth
##           -0.15751125           -0.21573234           -0.17545366
##         TeamBrentford          TeamBrighton           TeamBurnley
##           -0.23076241           -0.29305781           -0.57821575
##           TeamChelsea     TeamCrystal Palace           TeamEverton
##           -0.17749196           -0.52625760           -0.39037993
##            TeamFulham             TeamLeeds          TeamLiverpool
##           -0.38937612           -0.34255351           -0.09350578
##          TeamMan City         TeamMan United          TeamNewcastle
##            0.08940609           -0.00528952           -0.26486827
##     TeamNott'm Forest         TeamSunderland          TeamTottenham
##           -0.36827134           -0.50510090           -0.36221830
##          TeamWest Ham            TeamWolves    OpponentAston Villa
##           -0.39677870           -0.92783902            0.58228012
##    OpponentBournemouth      OpponentBrentford       OpponentBrighton
##            0.68176985            0.64086065            0.51490383
##       OpponentBurnley       OpponentChelsea OpponentCrystal Palace
##            0.99090690            0.64390942            0.60730912
##       OpponentEverton        OpponentFulham          OpponentLeeds
##            0.59346542            0.61331530            0.70910129
##     OpponentLiverpool      OpponentMan City     OpponentMan United
##            0.66812786            0.26618337            0.61581110
##     OpponentNewcastle  OpponentNott'm Forest     OpponentSunderland
##            0.69508633            0.61432022            0.54755773
##     OpponentTottenham      OpponentWest Ham         OpponentWolves
##            0.72583820            0.85552968            0.88158031
##                  Home
##            0.22099070
```

# The model

**Question.** Do you see any team missing?

```
length(unique(England2$Team))
```

```
## [1] 20
```

```
sum(startsWith(names(coefficients(model)), "Team"))
```

```
## [1] 19
```

There are 20 teams, but only 19 `Team` coefficients and 19 `Opponent` coefficients.

# The model

**Arsenal** is the missing team.

`glm` turns `Team` and `Opponent` into factors and drops the first level in alphabetical order. That reference level is Arsenal, so

- `TeamArsenal` $= 0$
- `OpponentArsenal` $= 0$

Both sit inside the intercept. Every other coefficient is a comparison with Arsenal.

In 2026 Arsenal finished **1st**, so almost every attack coefficient is negative: the other teams score less than Arsenal.

# The model: season 2020

In **2020** Arsenal finished **8th**. The same model still drops Arsenal, but now the teams that finished above Arsenal have a **positive** attack coefficient.

```
England_2020 <- f_data_sm %>%
  filter(SEASON == 2020, COUNTRY == "England") %>%
  select(HOMETEAM, AWAYTEAM, FTHG, FTAG)
eng1_2020 <- data.frame(England_2020[, c("HOMETEAM", "AWAYTEAM", "FTHG")], Home = 1)
eng2_2020 <- data.frame(England_2020[, c("AWAYTEAM", "HOMETEAM", "FTAG")], Home = 0)
colnames(eng1_2020) <- c("Team", "Opponent", "Goal", "Home")
colnames(eng2_2020) <- c("Team", "Opponent", "Goal", "Home")
model_2020 <- glm(Goal ~ Team + Opponent + Home,
                  data = rbind(eng1_2020, eng2_2020), family = poisson(link = log))
```

# The model: season 2020

Attack coefficients relative to Arsenal. A positive value means the team scores more than Arsenal.

```
attack_2020 <- coefficients(model_2020)
attack_2020 <- attack_2020[startsWith(names(attack_2020), "Team")]
sort(attack_2020, decreasing = TRUE)
```

```
##         TeamMan City        TeamLiverpool          TeamChelsea
##           0.58812419           0.40306440           0.21567765
##        TeamLeicester       TeamMan United        TeamTottenham
##           0.17266935           0.15246545           0.08474511
##      TeamSouthampton          TeamWolves         TeamWest Ham
##          -0.08146904          -0.10188795          -0.11953887
##          TeamEverton          TeamBurnley      TeamAston Villa
##          -0.23363573          -0.26277399          -0.29318028
##      TeamBournemouth         TeamBrighton TeamSheffield United
##          -0.31999622          -0.35657804          -0.37162972
##        TeamNewcastle          TeamWatford   TeamCrystal Palace
##          -0.37855766          -0.42664794          -0.59059184
##          TeamNorwich
##          -0.74157646
```

Man City, Liverpool, Chelsea, Leicester, Man United and Tottenham are above zero. Arsenal is still the zero.

# Making predictions

- Tottenham is playing against Man United at home

```
options(scipen=1, digits=4)
(tottenham_average <- predict(model, data.frame(Home=1, Team="Tottenham",
                                                Opponent="Man United"), type="respons
```

```
##     1
## 1.373
```

- The same for Man United

```
options(scipen=1, digits=4)
(mu_average <- predict(model, data.frame(Home=0, Team="Man United",
                                         Opponent="Tottenham"), type="response")
```

```
##     1
## 1.756
```

When Tottenham is playing Man United at Home, the expected (average) number of goals for Tottenham is 1.3731 and For Man United is 1.756

# Let's do the calculations by hand for Tottenham

## Calculations of $\lambda$ for Tottenham

$$log(\lambda) = \beta_0 + \beta_{TeamTottenham} + \beta_{OpponentManUnited} + \beta_{Home}$$

```
options(scipen=1, digits=4)
(intercept <- coefficients(model)["(Intercept)"])
```

```
## (Intercept)
##     -0.1575
```

```
(team_tottenham <- coefficients(model)["TeamTottenham"])
```

```
## TeamTottenham
##       -0.3622
```

```
(opponent_mu <- coefficients(model)["OpponentMan United"])
```

```
## OpponentMan United
##             0.6158
```

```
(home_coeff <- coefficients(model)["Home"])
```

```
##  Home
## 0.221
```

$log(\lambda) = 0.221 + \text{-}0.3622 + 0.6158 + \text{-}0.1575 = 0.3171$

$\lambda = e\hat{}(0.3171) = 1.3731$

# Making predictions: Home advantage

The average expected goals of the home team divided by the average expected goals by the same team playing away will be equal to exponent of Home coefficient

```
# lambda for Tottenham playing at home
tottenham_h=exp(intercept + team_tottenham + opponent_mu + home_coeff)
# lambda for Tottenham playing against Man United away
mu_a=exp(intercept + team_tottenham + opponent_mu)
tottenham_h/mu_a
```

```
## (Intercept)
##       1.247
```

```
exp(home_coeff)
```

```
##  Home
## 1.247
```

# Making predictions

As we have $\lambda$ for both teams, we can use skellam distribution to come up with the probabilities.

```
# Lisbon's lambda
tottenham_average
```

```
##     1
## 1.373
```

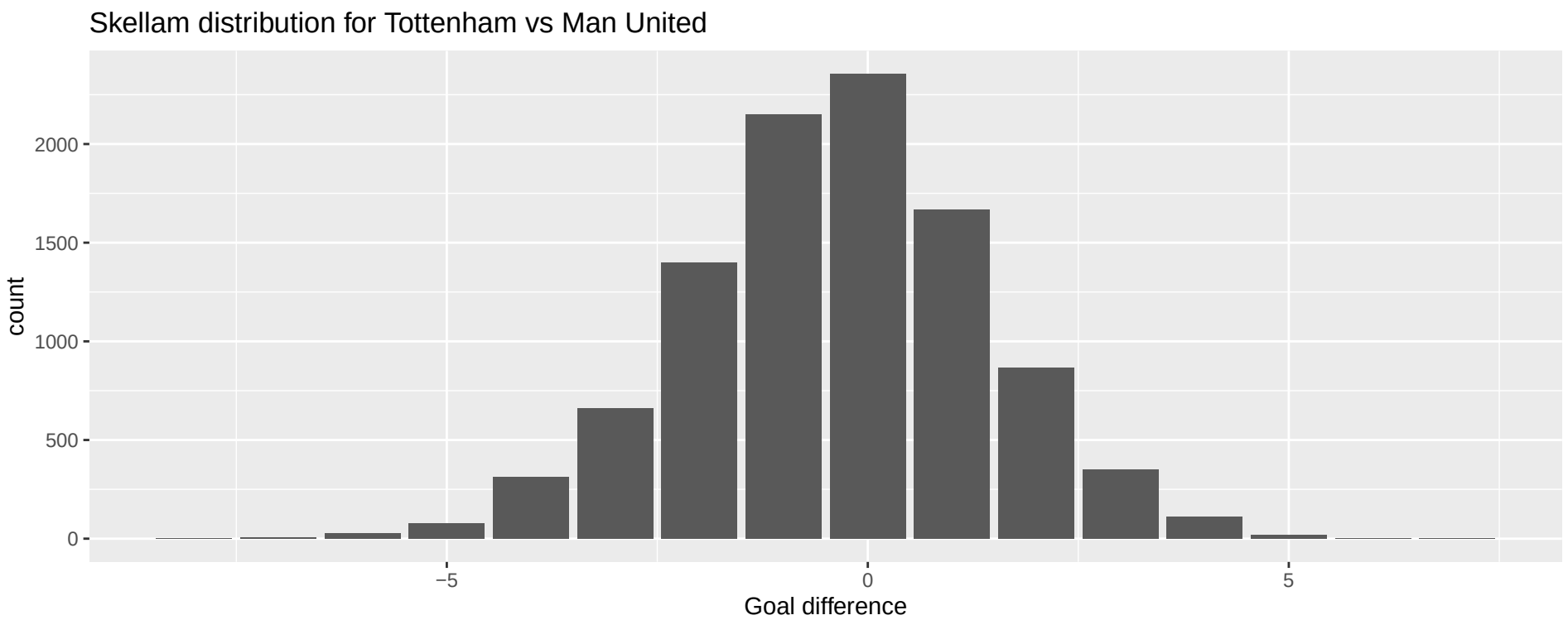
```
# Man United's lambda
mu_average
```

```
##     1
## 1.756
```

```
# Simulate game results
set.seed(1)
k = rskellam(10000,mu1 = tottenham_average, mu2 = mu_average)
```

# Making predictions

```
ggplot()+geom_bar(aes(x=k))+
  labs(x="Goal difference", title="Skellam distribution for Tottenham vs Man United"
```



# Making predictions

Calculate the game output probabilities

```
# Away team to win
sum(dskellam(c(-100:-1),tottenham_average, mu_average))
```

```
## [1] 0.4658
```

```
# Draw
sum(dskellam(0 ,tottenham_average, mu_average))
```

```
## [1] 0.2327
```

```
# Home team to win
sum(dskellam(c(1:100),tottenham_average, mu_average))
```

```
## [1] 0.3015
```

# Making predictions: Arsenal vs Man City

Arsenal is the reference team, so it has no coefficient of its own. Predict Arsenal at home against Man City. The actual score was 1-1.

```
(arsenal_average <- predict(model, data.frame(Home=1, Team="Arsenal",
                                              Opponent="Man City"), type="response"))
```

```
##    1
## 1.39
```

```
(city_average <- predict(model, data.frame(Home=0, Team="Man City",
                                           Opponent="Arsenal"), type="response"))
```

```
##      1
## 0.9342
```

# Making predictions: Arsenal vs Man City

The Arsenal terms are zero, so they drop out of the linear predictor.

$$log(\lambda_{Arsenal}) = \beta_0 + \beta_{OpponentManCity} + \beta_{Home}$$

$$log(\lambda_{ManCity}) = \beta_0 + \beta_{TeamManCity}$$

```
exp(intercept + opponent_city + home_coeff)
```

```
## (Intercept)
##        1.39
```

```
exp(intercept + team_city)
```

```
## (Intercept)
##      0.9342
```

# Relevel to the weakest team

The zero point is a choice. Wolves have the weakest attack in this fit.

Move the reference from Arsenal to Wolves. Predictions do not change, and neither does the home coefficient.

```
England3 <- England2 %>%
  mutate(Team = relevel(factor(Team), ref = "Wolves"),
         Opponent = relevel(factor(Opponent), ref = "Wolves"))
model_wolves <- glm(Goal ~ Team + Opponent + Home, data = England3,
                    family = poisson(link = log))
predict(model_wolves, data.frame(Home=1, Team="Arsenal", Opponent="Man City"),
        type="response")
```

```
##    1
## 1.39
```

# Compare the coefficients

Attack coefficients under the two reference teams. The missing team is filled in as 0.

```
teams <- c("Arsenal", "Man City", "Tottenham", "Wolves")
coef_or_zero <- function(mod, team) {
  val <- unname(coefficients(mod)[paste0("Team", team)])
  if (is.na(val)) 0 else val
}
vs_arsenal <- sapply(teams, coef_or_zero, mod = model)
vs_wolves <- sapply(teams, coef_or_zero, mod = model_wolves)
cbind(vs_Arsenal = vs_arsenal, vs_Wolves = vs_wolves,
      shift = vs_wolves - vs_arsenal)
```

```
##            vs_Arsenal vs_Wolves  shift
## Arsenal       0.00000    0.9278 0.9278
## Man City      0.08941    1.0172 0.9278
## Tottenham    -0.36222    0.5656 0.9278
## Wolves       -0.92784    0.0000 0.9278
```

The shift is the same for every team. Only the reference moved. The gap between two teams is unchanged.

# Team skills

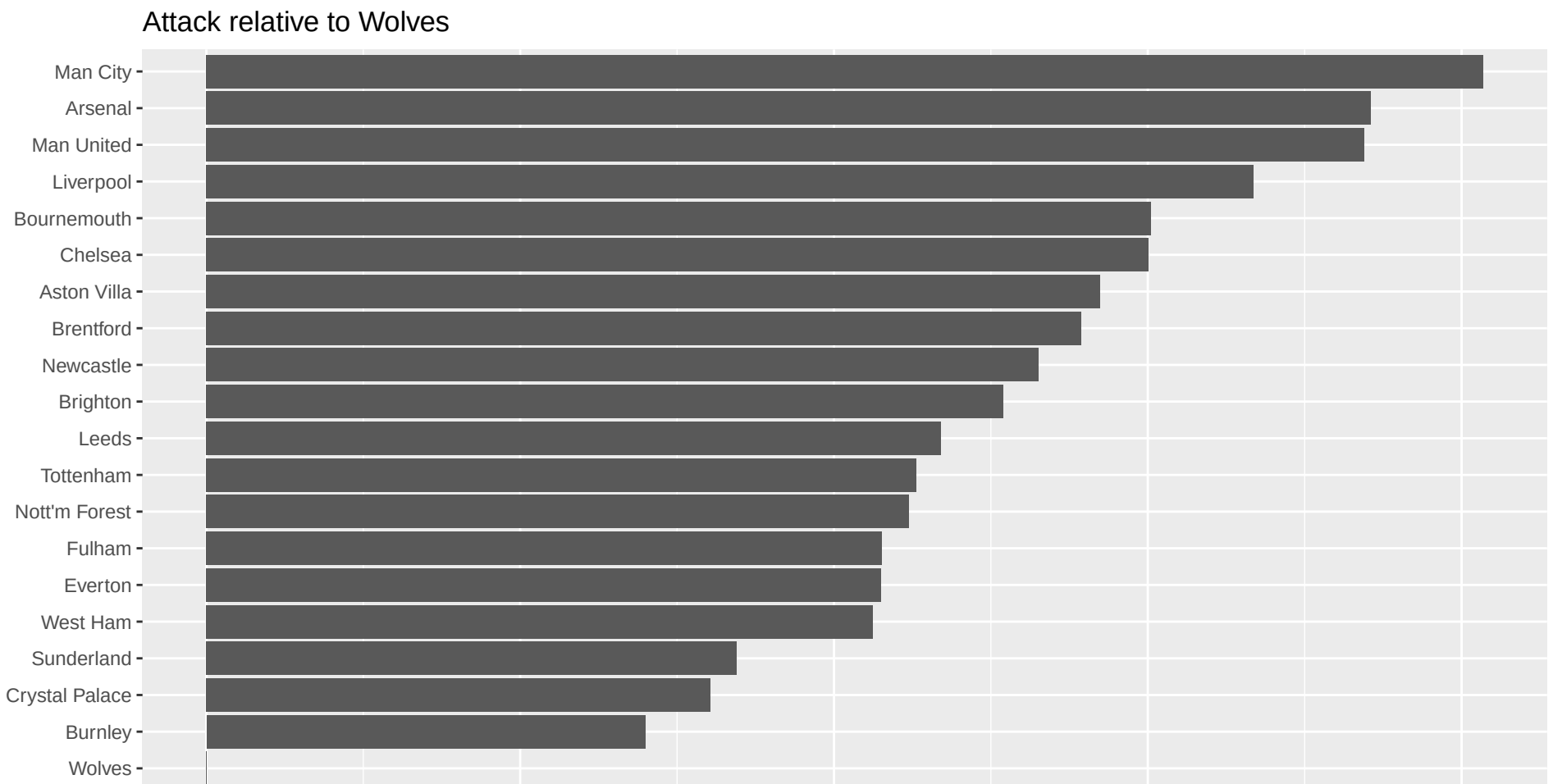
Attack is the `Team` coefficient. Defense is minus the `Opponent` coefficient, so a larger number is a stronger defense. Both are measured against Wolves.

```
teams <- levels(England3$Team)
attack <- sapply(teams, function(tm) {
  val <- coefficients(model_wolves)[paste0("Team", tm)]
  if (is.na(val)) 0 else unname(val)
})
defense <- sapply(teams, function(tm) {
  val <- coefficients(model_wolves)[paste0("Opponent", tm)]
  if (is.na(val)) 0 else -unname(val)
})
skills <- data.frame(team = teams,
                     attack = attack, defense = defense)
```

# Team skills

Attack relative to Wolves

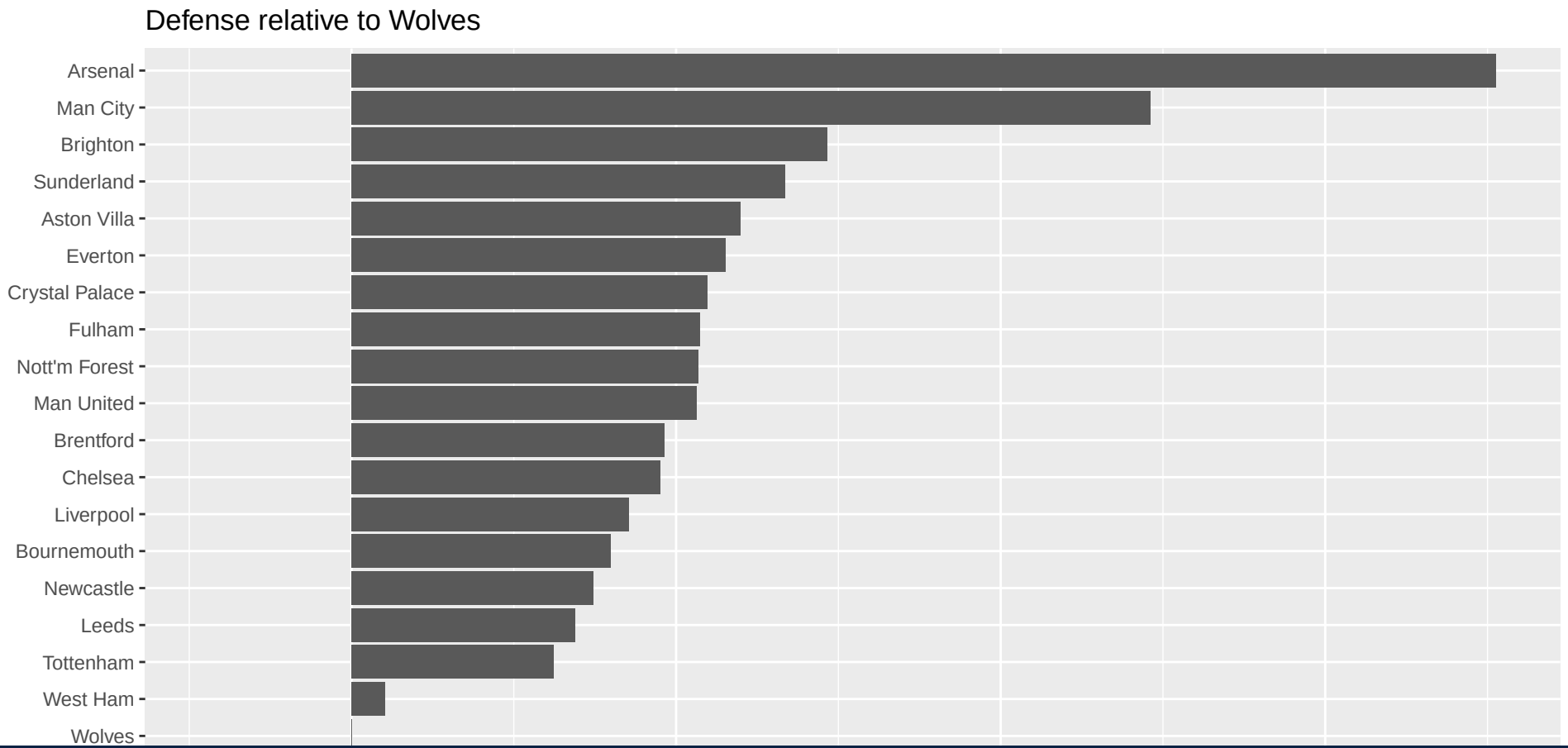
```
ggplot(skills, aes(x = reorder(team, attack), y = attack)) +
  geom_col() + coord_flip() +
  labs(x = NULL, y = "Attack coefficient", title = "Attack relative to Wolves")
```



# Team skills

Defense relative to Wolves. Wolves are the weakest attack, not the leakiest defense, so Burnley stays below zero.

```
ggplot(skills, aes(x = reorder(team, defense), y = defense)) +
  geom_col() + coord_flip() +
  labs(x = NULL, y = "Defense coefficient", title = "Defense relative to Wolves")
```



# Making predictions

**TIME FOR IN CLASS ACTIVITY !**

- Pick any game of your favorite team(or an arbitrary team) from last season.
  - It is best to pick a game that happened mid season to have some data from previous games as well.
  - Calculate the average number of goals scored by the home and away teams.
  - Calculate the average using only the games before the one you selected and in the same season.
  - Calculate the average using only the games before the one you selected and in the same season.
- Calculate the probabilities and odds of the home team winning, away team winning and draw using the distributions approach.
  - Use the matrix method.
- Calculate the expected score for the game using the Poisson Regression approach.
- Check the probability of that score according to the distributions approach.
- Compare the results with the actual game result.

# Predicting with models

## Bradley-Terry model

# Bradley-Terry model

The Bradley-Terry model (Bradley and Terry 1952) assumes that in a 'contest' between any two 'players', say player $i$ and player $j$ ( $i, j \in \{1, \ldots . , K\}$) the odds that $i$ beats $j$ are $\alpha_i/\alpha_j$ , where $\alpha_i$ and $\alpha_j$ are positive-valued parameters which represent the "ability" of each player/team.

The general model definition

$$P(\text{i beats j}) = \frac{\alpha_i}{\alpha_i + \alpha_j}$$

Sometimes the modification with exponential score function

$$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

# Bradley-Terry model

$$P(\text{i beats j}) = \frac{\alpha_i}{\alpha_i+\alpha_j}$$

$$logit(x) = \log(\frac{x}{1-x})$$

$$logit^{-1}(x) = \frac{1}{1+e^{(-x)}}$$

Log rules recap:

$$log(a*b) = log(a) + log(b)$$

$$log(a/b) = log(a) - log(b)$$

$$logit(P(\text{i beats j})) = \log(\frac{P(\text{i beats j})}{1-P(\text{i beats j})}) = \log(\frac{P(\text{i beats j})}{P(\text{j beats i})}) = log(\frac{\alpha_i}{\alpha_i+\alpha_j} * \frac{1}{\frac{\alpha_j}{\alpha_i+\alpha_j}}) =$$

$$log(\frac{\alpha_i}{\alpha_i+\alpha_j} * \frac{\alpha_i+\alpha_j}{\alpha_j}) = log(\frac{\alpha_i}{\alpha_i+\alpha_j}) + log(\frac{\alpha_i+\alpha_j}{\alpha_j}) =$$

$$log(\alpha_i) - log(\alpha_i+\alpha_j) + log(\alpha_i+\alpha_j) - log(\alpha_j) = log(\alpha_i) - log(\alpha_j) = \lambda_i - \lambda_j$$

The goal of the modeling is to find the best estimate for $\lambda_i, \lambda_j$

# Bradley-Terry model

The modification with exponential score function

$$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

$$logit(P(\text{i beats j})) = \log(\frac{P(\text{i beats j})}{1 - P(\text{i beats j})}) = \log(\frac{P(\text{i beats j})}{P(\text{j beats i})}) = log(\frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}} * \frac{e^{\alpha_i} + e^{\alpha_j}}{e^{\alpha_j}}) = \alpha_i - \alpha_j$$

logit - log of odds

# Bradley-Terry model

The basic Bradley-Terry model does not predict draw/tie games.

You can use Davidson's extension on Bradley-Terry model if want to predict draw:

https://www.jstor.org/stable/2283595?seq=1#page_scan_tab_contents

# Bradley-Terry model

```
data(nfl)
```

```
head(nfl, n=5)
```

```
##   schedule_season schedule_week schedule_date           team_away score_away
## 1            2000             1    2000-09-03  San Francisco 49ers         28
## 2            2000             1    2000-09-03     Tennessee Titans         13
## 3            2000             1    2000-09-03 Jacksonville Jaguars         27
## 4            2000             1    2000-09-03  Philadelphia Eagles         41
## 5            2000             1    2000-09-03        New York Jets         20
##           team_home score_home              stadium stadium_type
## 1   Atlanta Falcons         36         Georgia Dome       indoor
## 2     Buffalo Bills         16 Ralph Wilson Stadium      outdoor
## 3  Cleveland Browns          7  FirstEnergy Stadium      outdoor
## 4    Dallas Cowboys         14        Texas Stadium      outdoor
## 5 Green Bay Packers         16        Lambeau Field      outdoor
##   stadium_location result spread home_odds_open away_odds_open home_odds_close
## 1      Atlanta, GA      H     -8           <NA>           <NA>            <NA>
## 2 Orchard Park, NY      H     -3           <NA>           <NA>            <NA>
## 3    Cleveland, OH      A     20           <NA>           <NA>            <NA>
## 4       Irving, TX      A     27           <NA>           <NA>            <NA>
## 5    Green Bay, WI      A      4           <NA>           <NA>            <NA>
##   away_odds_close
## 1            <NA>
## 2            <NA>
## 3            <NA>
## 4            <NA>
## 5            <NA>
```

# Bradley-Terry model

- Data on NFL matches
- Description
- Format
- A data frame with 4848 rows and 16 variables
- schedule_season: from 2000-2025
- schedule_week: The number of the week when the game was played
- schedule_date: The actual date of the game
- team_away: Away team
- score_away: The score of the away team
- team_home: Home team
- score_home: The score of home team
- stadium: The name of the stadium
- stadium_type: indoor, outdoor, retractable
- stadium_location: The location of the stadium: City and state abbreviation
- odds: closing and opening betting odds
- result: The result of the game: H=Home Win, T=Tie, A=Away Win
- spread: points spread, the difference between away team and home team scores ,
- home_odds_open, home_odds_close: Opening and closing decimal odds for the home team
- home_odds_open, away_odds_close: Opening and closing decimal odds for the away team

# Bradley-Terry model

Some data preparation

```
nfl2 = nfl %>%
  select(team_home, team_away, result) %>%
  filter(result !="T") %>%
  mutate(ht_w = ifelse(result=="H",1,0),
         at_w = ifelse(result=="A",1,0))
head(nfl2)
```

```
##            team_home            team_away result ht_w at_w
## 1    Atlanta Falcons  San Francisco 49ers      H    1    0
## 2      Buffalo Bills     Tennessee Titans      H    1    0
## 3   Cleveland Browns Jacksonville Jaguars      A    0    1
## 4     Dallas Cowboys  Philadelphia Eagles      A    0    1
## 5  Green Bay Packers        New York Jets      A    0    1
## 6 Kansas City Chiefs   Indianapolis Colts      A    0    1
```

# Bradley-Terry model

The final dataset

```
nfl3 = nfl2 %>%
    mutate(team_home=as.factor(team_home),
           team_away = as.factor(team_away)) %>%
    group_by(team_home, team_away) %>%
    summarise(ht=sum(ht_w), at=sum(at_w))

head(nfl3, n=5)
```

```
## # A tibble: 5 x 4
## # Groups:   team_home [1]
##   team_home          team_away            ht    at
##   <fct>              <fct>             <dbl> <dbl>
## 1 Arizona Cardinals Atlanta Falcons       5     2
## 2 Arizona Cardinals Baltimore Ravens      1     4
## 3 Arizona Cardinals Buffalo Bills         2     1
## 4 Arizona Cardinals Carolina Panthers     3     6
## 5 Arizona Cardinals Chicago Bears         1     3
```

# Bradley-Terry model

Fitting the bradley-terry model

```
install.packages("BradleyTerry2")
```

```
outcome_matrix <- cbind(nfl3$ht, nfl3$at)
model = BTm(outcome=outcome_matrix, player1=team_home,
            player2=team_away, data=nfl3, id="team_")
```

# Bradley-Terry model

Coefficients represent the teams abilities to win

```
summary(model)
```

```
##
## Call:
## BTm(outcome = outcome_matrix, player1 = team_home, player2 = team_away,
##     id = "team_", data = nfl3)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## team_Atlanta Falcons         0.2912     0.1299    2.24  0.02503 *
## team_Baltimore Ravens        0.8386     0.1346    6.23  4.6e-10 ***
## team_Buffalo Bills           0.5926     0.1340    4.42  9.8e-06 ***
## team_Carolina Panthers       0.1248     0.1298    0.96  0.33625
## team_Chicago Bears           0.1842     0.1315    1.40  0.16123
## team_Cincinnati Bengals      0.4638     0.1337    3.47  0.00052 ***
## team_Cleveland Browns       -0.0648     0.1352   -0.48  0.63178
## team_Dallas Cowboys          0.6502     0.1309    4.97  6.7e-07 ***
## team_Denver Broncos          0.5051     0.1324    3.82  0.00014 ***
## team_Detroit Lions           0.1271     0.1314    0.97  0.33341
## team_Green Bay Packers       0.8007     0.1326    6.04  1.6e-09 ***
## team_Houston Texans          0.0900     0.1359    0.66  0.50818
## team_Indianapolis Colts      0.5748     0.1337    4.30  1.7e-05 ***
## team_Jacksonville Jaguars    0.0460     0.1339    0.34  0.73137
## team_Kansas City Chiefs      0.7901     0.1338    5.91  3.5e-09 ***
## team_Las Vegas Raiders       0.0217     0.1337    0.16  0.87127
## team_Los Angeles Chargers    0.4688     0.1323    3.54  0.00039 ***
## team_Los Angeles Rams        0.3028     0.1239    2.44  0.01455 *
## team_Miami Dolphins          0.4539     0.1334    3.40  0.00066 ***
## team_Minnesota Vikings       0.6090     0.1311    4.65  3.4e-06 ***
## team_New England Patriots    0.9964     0.1351    7.37  1.7e-13 ***
## team_New Orleans Saints      0.4968     0.1305    3.81  0.00014 ***
## team_New York Giants         0.2899     0.1308    2.22  0.02663 *
## team_New York Jets           0.1623     0.1341    1.21  0.22618
## team_Philadelphia Eagles     0.9014     0.1321    6.82  9.0e-12 ***
```

# Bradley-Terry model

New England Patriots: the strongest team  Cleveland Browns: the weakest team

```
coef=model$coefficients
sort(coef, decreasing = T)
```

```
##  team_New England Patriots   team_Pittsburgh Steelers
##                    0.99641                    0.91669
##   team_Philadelphia Eagles      team_Baltimore Ravens
##                    0.90135                    0.83864
##     team_Green Bay Packers    team_Kansas City Chiefs
##                    0.80065                    0.79010
##      team_Seattle Seahawks       team_Dallas Cowboys
##                    0.66231                    0.65024
##     team_Minnesota Vikings        team_Buffalo Bills
##                    0.60901                    0.59258
##    team_Indianapolis Colts       team_Denver Broncos
##                    0.57479                    0.50510
##   team_San Francisco 49ers    team_New Orleans Saints
##                    0.49807                    0.49684
##  team_Los Angeles Chargers    team_Cincinnati Bengals
##                    0.46884                    0.46378
##        team_Miami Dolphins      team_Los Angeles Rams
##                    0.45393                    0.30284
##       team_Atlanta Falcons       team_New York Giants
##                    0.29119                    0.28990
##  team_Tampa Bay Buccaneers      team_Tennessee Titans
##                    0.27228                    0.25539
##         team_Chicago Bears        team_New York Jets
##                    0.18417                    0.16226
## team_Washington Commanders        team_Detroit Lions
##                    0.13651                    0.12712
##     team_Carolina Panthers       team_Houston Texans
##                    0.12481                    0.08995
##  team_Jacksonville Jaguars     team_Las Vegas Raiders
##                    0.04596                    0.02167
##      team_Cleveland Browns
```

# Bradley-Terry model

Overall there are 32 teams in the data

```
length(unique(nfl3$team_home))
```

```
## [1] 32
```

```
# and we have coefficients for 31
length(coef)
```

```
## [1] 31
```

- One team is taken as a reference level
- Its ability is $= 0$
- In our case that is Arizona Cardinals as it comes first alphabetically

# Bradley-Terry model

- Abilities are calculated relative to Arizona Cardinales
- Thus, positive ability means that the team is doing better than Cardinals
- Negative ability is showing that the team is doing worst than Cardinals

```
BTabilities(model)
```

```
##                         ability   s.e.
## Arizona Cardinals       0.00000 0.0000
## Atlanta Falcons         0.29119 0.1299
## Baltimore Ravens        0.83864 0.1346
## Buffalo Bills           0.59258 0.1340
## Carolina Panthers       0.12481 0.1298
## Chicago Bears           0.18417 0.1315
## Cincinnati Bengals      0.46378 0.1337
## Cleveland Browns       -0.06477 0.1352
## Dallas Cowboys          0.65024 0.1309
## Denver Broncos          0.50510 0.1324
## Detroit Lions           0.12712 0.1314
## Green Bay Packers       0.80065 0.1326
## Houston Texans          0.08995 0.1359
## Indianapolis Colts      0.57479 0.1337
## Jacksonville Jaguars    0.04596 0.1339
## Kansas City Chiefs      0.79010 0.1338
## Las Vegas Raiders       0.02167 0.1337
```

# Bradley-Terry model

- Let's do some plotting
- First let's create a dataframe with abilities

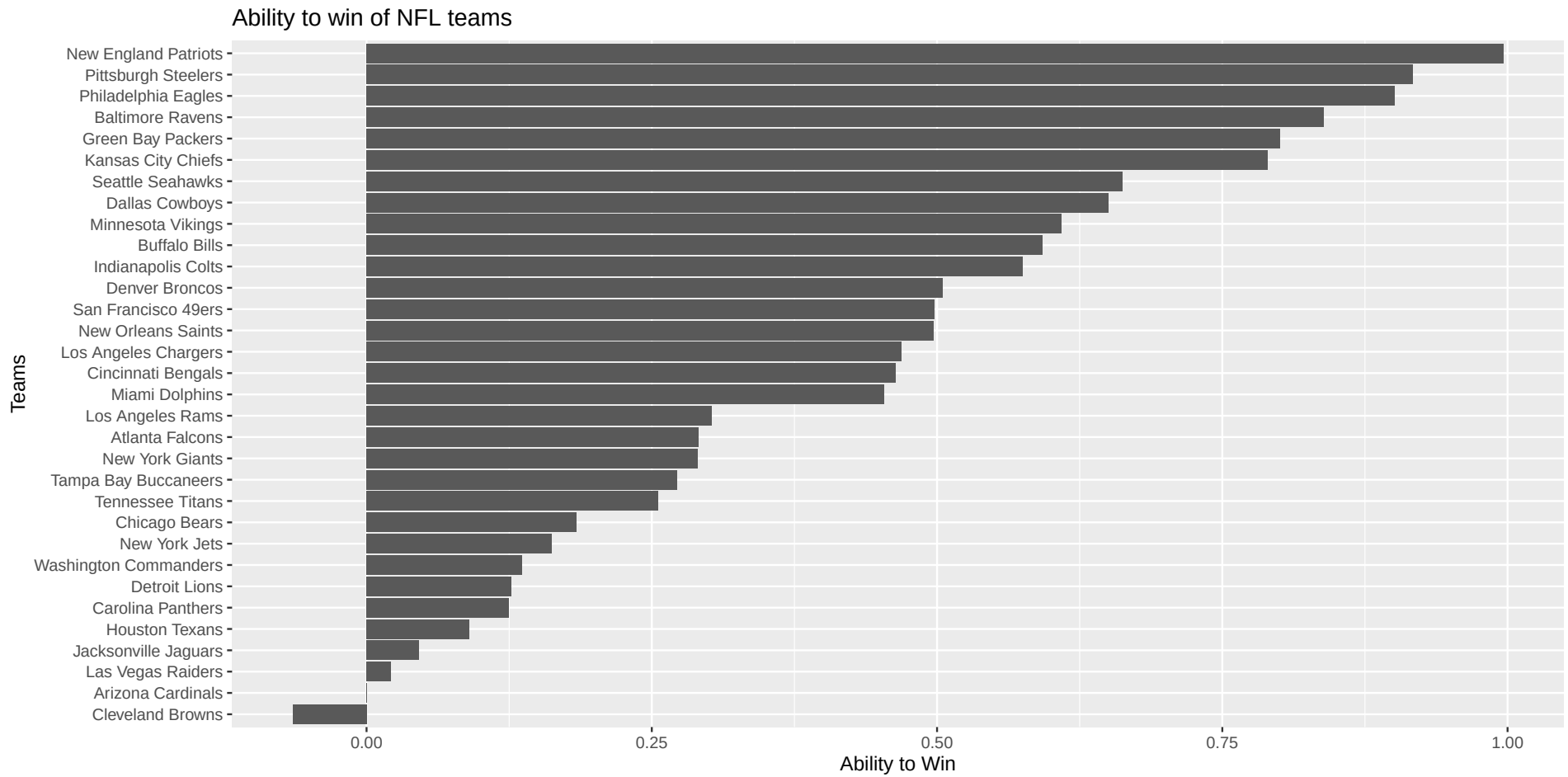
```
abilities = as.data.frame(BTabilities(model))
abilities$team = rownames(abilities)
abilities = abilities[order(abilities$ability, decreasing = T),]
abilities
```

```
##                         ability   s.e.                    team
## New England Patriots    0.99641 0.1351  New England Patriots
## Pittsburgh Steelers     0.91669 0.1353   Pittsburgh Steelers
## Philadelphia Eagles     0.90135 0.1321   Philadelphia Eagles
## Baltimore Ravens        0.83864 0.1346      Baltimore Ravens
## Green Bay Packers       0.80065 0.1326     Green Bay Packers
## Kansas City Chiefs      0.79010 0.1338    Kansas City Chiefs
## Seattle Seahawks        0.66231 0.1248      Seattle Seahawks
## Dallas Cowboys          0.65024 0.1309        Dallas Cowboys
## Minnesota Vikings       0.60901 0.1311     Minnesota Vikings
## Buffalo Bills           0.59258 0.1340         Buffalo Bills
## Indianapolis Colts      0.57479 0.1337    Indianapolis Colts
## Denver Broncos          0.50510 0.1324        Denver Broncos
## San Francisco 49ers     0.49807 0.1239   San Francisco 49ers
## New Orleans Saints      0.49684 0.1305    New Orleans Saints
## Los Angeles Chargers    0.46884 0.1323  Los Angeles Chargers
```

# Bradley-Terry model

Abilities from strongest to weakest

```
ggplot(data=abilities, aes(x = reorder(team, ability), y = ability)) +
  geom_bar(stat="identity") + coord_flip() +
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



# Bradley-Terry model

Everything relative to the worst team (Cleveland Browns)

- Let's say we want to calculate the abilities relative to the weakest team, so all the abilities would be $\geq 0$
- You could do this by re leveling the team variables

```
nfl4=nfl2 %>%
    mutate(team_home = relevel(team_home, ref="Cleveland Browns"),
           team_away = relevel(team_away, ref="Cleveland Browns")) %>%
    group_by(team_home, team_away) %>% summarise(ht=sum(ht_w), at=sum(at_w))
model_browns = BTm(cbind(ht, at), team_home, team_away, data=nfl4, id="team_")
```

Standardize the variable ability around its minimum (these two approaches will give exactly the same result)

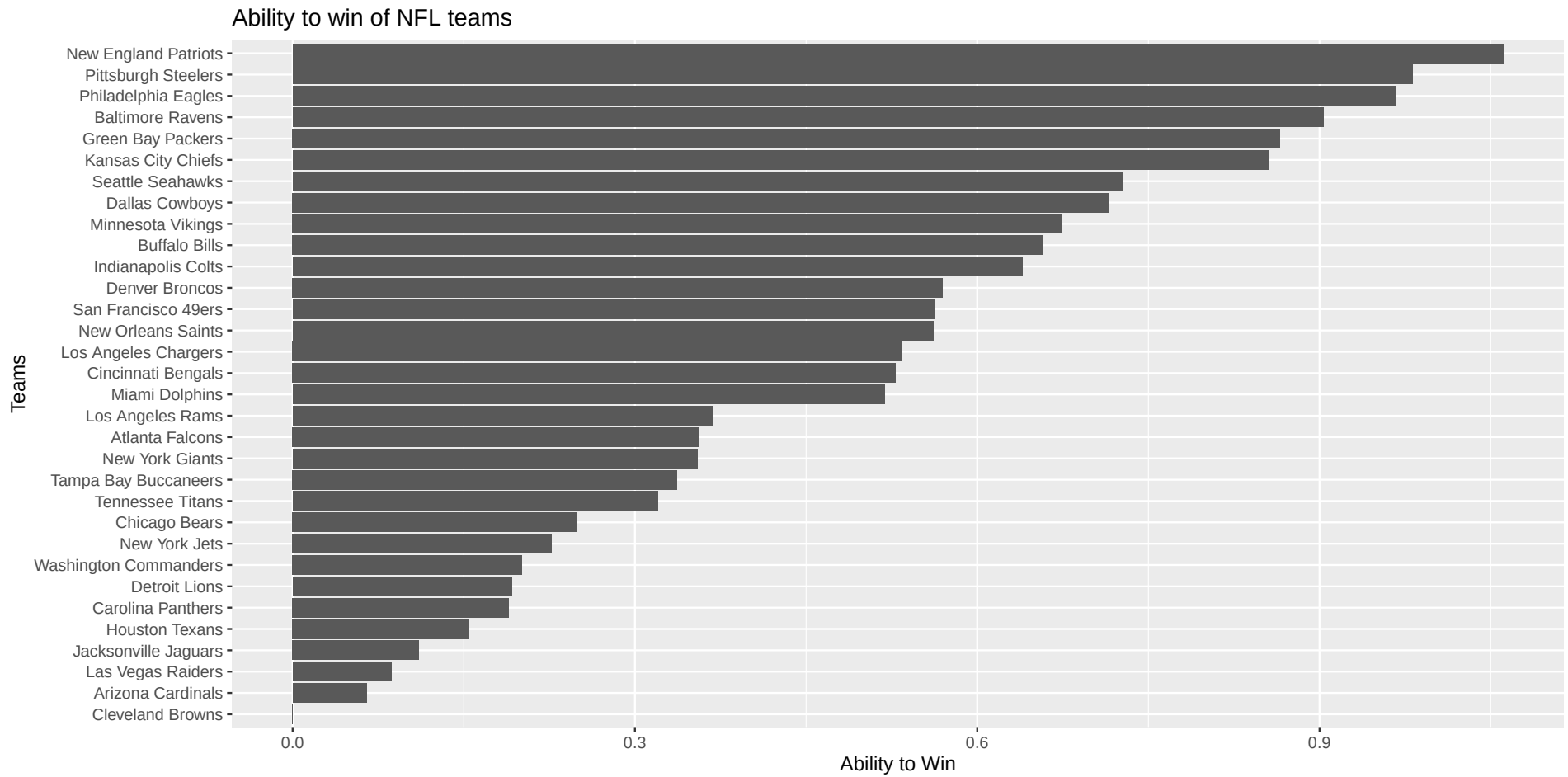
```
abilities_brown = abilities
abilities_brown$ability = abilities_brown$ability - min(abilities$ability)
head(abilities_brown)
```

```
##                      ability   s.e.                 team
## New England Patriots  1.0612 0.1351 New England Patriots
## Pittsburgh Steelers   0.9815 0.1353  Pittsburgh Steelers
## Philadelphia Eagles   0.9661 0.1321  Philadelphia Eagles
## Baltimore Ravens      0.9034 0.1346     Baltimore Ravens
## Green Bay Packers     0.8654 0.1326    Green Bay Packers
## Kansas City Chiefs    0.8549 0.1338   Kansas City Chiefs
```

# Bradley-Terry model

## Abilities plot

```
ggplot(data=abilities_brown, aes(x=reorder(team,ability), y=ability)) +
  geom_bar(stat="identity") + coord_flip() +
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



# Bradley-Terry model

Let's do predictions for a sample of games for Denver Broncos

```
broncos=data.frame(team_home=rep("Denver Broncos", 4),
                   team_away=c("Cleveland Browns", "Tennessee Titans",
                               "Chicago Bears", "Detroit Lions"))
# we need to make team_home and team_away having the same factor structure
# as in the initial model

broncos$team_home=factor(broncos$team_home,
                         levels(nfl3$team_away))

broncos$team_away=factor(broncos$team_away,
                         levels(nfl3$team_away))
str(broncos)
```

```
## 'data.frame':    4 obs. of  2 variables:
##  $ team_home: Factor w/ 32 levels "Arizona Cardinals",..: 10 10 10 10
##  $ team_away: Factor w/ 32 levels "Arizona Cardinals",..: 8 31 6 11
```

```
broncos
```

```
##        team_home        team_away
## 1 Denver Broncos Cleveland Browns
## 2 Denver Broncos Tennessee Titans
## 3 Denver Broncos    Chicago Bears
## 4 Denver Broncos    Detroit Lions
```

# Bradley-Terry model

This is the probability of Denver Broncos winning

```
( broncos_prob = predict(model, newdata=broncos, level=2, type="response") )
```

```
##      1      2      3      4
## 0.6387 0.5621 0.5795 0.5934
```

Make a dataframe with probs

```
( broncos_df = data.frame(broncos, ht_w=broncos_prob,
                                   at_w=1-broncos_prob) )
```

```
##        team_home        team_away   ht_w   at_w
## 1 Denver Broncos Cleveland Browns 0.6387 0.3613
## 2 Denver Broncos Tennessee Titans 0.5621 0.4379
## 3 Denver Broncos    Chicago Bears 0.5795 0.4205
## 4 Denver Broncos    Detroit Lions 0.5934 0.4066
```

# Bradley-Terry model

Probabilities are predicted using the following formula

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$

$P(\text{j beats i}) = \frac{e^{\alpha_j}}{e^{\alpha_i} + e^{\alpha_j}}$

# Bradley-Terry model

Calculating probabilities manually for Denver Broncos-Cleveland Browns

```
#first get abbilities
broncos_ab = abilities[rownames(abilities)=="Denver Broncos",1]
browns_ab = abilities[rownames(abilities)=="Cleveland Browns",1]

broncos_ab
```

```
## [1] 0.5051
```

```
browns_ab
```

```
## [1] -0.06477
```

```
# probability of Broncos to win
exp(broncos_ab)/(exp(broncos_ab)+exp(browns_ab))
```

```
## [1] 0.6387
```

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i}+e^{\alpha_j}}$

# Bradley-Terry model with order effect

## Bradley-Terry model with order effect

# Bradley-Terry model with order effect

- In certain types of application some or all contests have an associated 'bias', related to the order in which items are presented to a judge or with the location in which a contest takes place, for example.
- In sports this can be related to home advantage: the ability of the team is different when it plays at home compared to when it plays away.
- In this case the model has the following structure

$$logit(P(\text{i beats j})) = \lambda_i - \lambda_j + \beta * z$$

where

z = 1 if i has the supposed advantage and.

z = -1 if j has it.

z = 0 if there is no advantage.

$\beta$ is the coefficient of the advantage.

# Bradley-Terry model with order effect

We need to do several data transformations to get the model to account home advantage factor.

```
nfl3$team_home = data.frame(team = nfl3$team_home, at.home = 1)
nfl3$team_away = data.frame(team = nfl3$team_away, at.home = 0)

nfl3
```

```
## # A tibble: 992 x 4
## # Groups:   team_home [32]
##    team_home$team    $at.home team_away$team     $at.home    ht    at
##    <fct>                <dbl> <fct>                 <dbl> <dbl> <dbl>
##  1 Arizona Cardinals        1 Atlanta Falcons           0     5     2
##  2 Arizona Cardinals        1 Baltimore Ravens          0     1     4
##  3 Arizona Cardinals        1 Buffalo Bills             0     2     1
##  4 Arizona Cardinals        1 Carolina Panthers         0     3     6
##  5 Arizona Cardinals        1 Chicago Bears             0     1     3
##  6 Arizona Cardinals        1 Cincinnati Bengals        0     2     2
##  7 Arizona Cardinals        1 Cleveland Browns          0     4     0
##  8 Arizona Cardinals        1 Dallas Cowboys            0     8     2
##  9 Arizona Cardinals        1 Denver Broncos            0     1     3
## 10 Arizona Cardinals        1 Detroit Lions             0     7     3
## # i 982 more rows
```

The result is a grouped_df

# Bradley-Terry model with order effect

```
head(nfl3$team_home, n=5)
```

```
##                 team at.home
## 1 Arizona Cardinals       1
## 2 Arizona Cardinals       1
## 3 Arizona Cardinals       1
## 4 Arizona Cardinals       1
## 5 Arizona Cardinals       1
```

# Bradley-Terry model with order effect

```
model2 = BTm(cbind(ht, at), team_home, team_away, formula=~team+at.home,
             data=nfl3, id="team")
```

```
model2$coefficients
```

```
##       teamAtlanta Falcons      teamBaltimore Ravens         teamBuffalo Bills
##                   0.29333                   0.85153                   0.59814
##     teamCarolina Panthers        teamChicago Bears    teamCincinnati Bengals
##                   0.12479                   0.18340                   0.47393
##      teamCleveland Browns        teamDallas Cowboys        teamDenver Broncos
##                  -0.06491                   0.65787                   0.51197
##         teamDetroit Lions     teamGreen Bay Packers        teamHouston Texans
##                   0.12422                   0.81017                   0.09395
##    teamIndianapolis Colts  teamJacksonville Jaguars    teamKansas City Chiefs
##                   0.58159                   0.04605                   0.80292
##     teamLas Vegas Raiders  teamLos Angeles Chargers      teamLos Angeles Rams
##                   0.02149                   0.47593                   0.30751
##        teamMiami Dolphins     teamMinnesota Vikings  teamNew England Patriots
##                   0.46028                   0.61498                   1.00913
##    teamNew Orleans Saints       teamNew York Giants        teamNew York Jets
##                   0.50327                   0.29285                   0.16358
##   teamPhiladelphia Eagles   teamPittsburgh Steelers   teamSan Francisco 49ers
##                   0.91195                   0.93132                   0.50637
##      teamSeattle Seahawks  teamTampa Bay Buccaneers      teamTennessee Titans
##                   0.67006                   0.27514                   0.25885
## teamWashington Commanders                   at.home
##                   0.13365                   0.25289
```

# Bradley-Terry model with order effect

Let make predictions for Broncos vs Browns by hand, assuming that Broncos are playing at home

```
( db = coefficients(model2)['teamDenver Broncos'] )
```

```
## teamDenver Broncos
##              0.512
```

```
( cb = coefficients(model2)['teamCleveland Browns'] )
```

```
## teamCleveland Browns
##             -0.06491
```

```
(h = coefficients(model2)['at.home'])
```

```
## at.home
##  0.2529
```

# Bradley-Terry model with order effect

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$

Probability of Broncos to win at home

```
( w_h = exp(db+h)/(exp(db+h)+exp(cb)) )
```

```
## teamDenver Broncos
##             0.6963
```

Odds of Broncos to win at home

```
( odd_h = w_h/(1-w_h) )
```

```
## teamDenver Broncos
##              2.293
```

# Bradley-Terry model with order effect

Playing at "neutral" field, no advantage for any team

$P(\text{j beats j}) = \frac{e^{\alpha_j}}{e^{\alpha_i}+e^{\alpha_j}}$

```
( w_n = exp(db)/(exp(db)+exp(cb)) )
```

```
## teamDenver Broncos
##             0.6403
```

Odds of wining at "neutral" field

```
( odd_n = w_n/(1-w_n) )
```

```
## teamDenver Broncos
##               1.78
```

# Bradley-Terry model with order effect

Odds ratio and home team advantage

```
(odd_h/odd_n)
```

```
## teamDenver Broncos
##              1.288
```

```
(exp(coefficients(model2)['at.home']))
```

```
## at.home
##   1.288
```

$e^{\text{at.home}}$ is the odd multiplier for a home team to win compared to the neutral field

# Bradley-Terry model with order effect

The same game, now Broncos is away, Browns is at home

$P(\text{j beats i}) = \frac{e^{\alpha_j + h}}{e^{\alpha_i} + e^{\alpha_j + h}}$

```
( w_a = exp(db-h)/(exp(db-h)+exp(cb)) )
```

```
## teamDenver Broncos
##             0.5803
```

Probability of Browns to win at home

```
1 - exp(db-h)/(exp(db-h)+exp(cb))
```

```
## teamDenver Broncos
##             0.4197
```

```
# OR
exp(cb+h)/(exp(db)+exp(cb+h))
```

```
## teamCleveland Browns
##             0.4197
```

# Bradley-Terry model

Making predictions for the series of games

```
broncos = data.frame(team_home=rep("Denver Broncos", 4),
                     team_away=c("Cleveland Browns", "Tennessee Titans",
                                 "Chicago Bears", "Detroit Lions"))

broncos$team_home=data.frame(team = factor(broncos$team_home,
                                           levels(nfl3$team_home$team)),
                                           at.home = 1)
broncos$team_away=data.frame(team = factor(broncos$team_away,
                                           levels(nfl3$team_home$team)),
                                           at.home = 0)
broncos
```

```
##   team_home.team team_home.at.home   team_away.team team_away.at.home
## 1 Denver Broncos                 1 Cleveland Browns                 0
## 2 Denver Broncos                 1 Tennessee Titans                 0
## 3 Denver Broncos                 1    Chicago Bears                 0
## 4 Denver Broncos                 1    Detroit Lions                 0
```

# Bradley-Terry model

```
( br2 = predict(model2, newdata=broncos, level=2, type="response") )
```

```
##      1      2      3      4
## 0.6963 0.6239 0.6414 0.6549
```